

QRCx: Sequential Dissipative Quantum Reservoir Computing for Operational Weather Forecasting

QRCx Team

qBraid $\times$ MITRE $\times$ JonesTrading Global Industry Challenge 2026, Phase 3, Track B

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1 Track and Rationale

We build a sequential dissipative quantum reservoir computer (QRC) for real hourly weather forecasting at NOAA station KORD (Chicago O’Hare), and hold it to a fairness protocol no reviewer can attack: identical features, identical strict temporal splits, and a dimension-matched classical echo-state network (ESN) plus a null-control linear baseline evaluated under the exact same protocol. Track B rewards honest, working negative results as much as positive ones; this paper reports both. Our core scientific question is narrow and falsifiable: *does engineered dissipation, not just unitary evolution, provide the reservoir’s forecasting-relevant memory?* We answer yes, with a statistically significant ablation (Sec. 4.2), while also reporting honestly that the resulting model does not yet beat a simple linear baseline on this task (Sec. 4.1) – a finding the fairness protocol was specifically built to surface, not hide.

2 Architecture

Figure 1 shows the pipeline. A real hourly 13-feature vector x_k (dry-bulb and dew-point temperature, relative humidity, wind speed/direction decomposed into W_x, W_y , sea-level pressure, temperature depression, and four cyclical hour-of-day/day-of-year encodings) is injected via single-qubit RY rotations (or, in an alternative encoding mode, a ZZ-feature-map-style layer) into a transverse-field Ising model (TFIM) reservoir of N qubits, whose density matrix ρ_k evolves coherently under the exact propagator $e^{-iH\tau}$ (eigendecomposition-based, not Trotterized) before a dissipative channel D_{γ_1, γ_2} – amplitude damping (γ_1) and dephasing (γ_2) – is applied, and *persists to the next timestep*: this is a genuinely recurrent reservoir, not a fresh-encode-per-sample windowed one. A time-multiplexed readout takes V equally-spaced intermediate snapshots of single- and two-body Pauli correlators per coherent-evolution interval (non-destructively; ρ keeps evolving), giving $3N + 3\binom{N}{2}$ correlators per sub-step (234 at the $N = 12$ reference configuration). A Ridge regression readout predicts the climatological-anomaly residual target, either directly or as a delta added back to persistence (“residual” architecture). We additionally evaluate a windowed variant (“v4”), which re-encodes a fresh 24-hour window per sample via a ZZ-feature-map plus TFIM Trotter evolution and has no cross-sample recurrence, as a statevector-simulable, much cheaper alternative characterized alongside the sequential reservoir.

ρ_k carries forward to step $k+1$ (recurrent state)

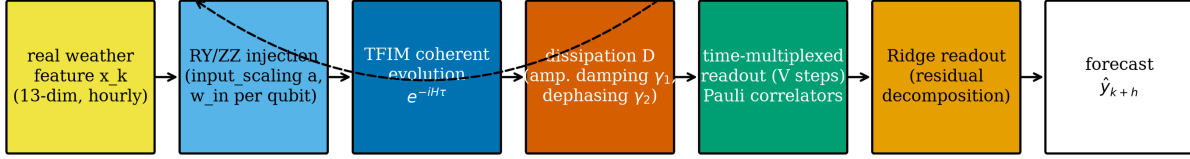


Figure 1: Sequential dissipative reservoir pipeline. ρ_k (dashed feedback) carries the recurrent quantum state from step k to $k + 1$ – the architectural feature this paper’s central ablation (Sec. 4.2) isolates.

3 IPC-Matched Tuning (Novel Contribution)

Reservoir hyperparameters are conventionally tuned to maximize *total* Information Processing Capacity (IPC; Dambre et al. 2012, extended to dissipative reservoirs by Čindrak et al. 2026) – but a forecasting task only benefits from capacity in the specific (delay, degree) region it actually uses. We introduce a **task demand profile** $D(\text{delay}, \text{degree})$: how much of the real KORD residual target’s h -step-ahead value is explained by a degree-1/2/3 Legendre polynomial of its own lagged history, at the same (delay, degree) resolution as the reservoir’s supply-side IPC $C(\text{lag}, \text{degree})$ (extended here to degree 3 via probabilists’ Hermite polynomials of the reservoir’s iid-Gaussian drive). Matching – maximizing captured capacity $\sum_h \sum \min(C, D)$ over a coarse (γ_1, a) grid rather than raw $\sum C$ – selected $\gamma_1=0.3, a=0.1$ (captured capacity 9.26, vs. 4.53 for the prior reference config: a +104% overlap improvement). **Honest result:** on a real (reduced-sample) pilot forecast comparison, the matched configuration improved skill at $h=1, 3$ but the reference configuration remained better at $h=6$ (this task’s own named target horizon) and $h=12$ (Table 1) – maximizing univariate self-demand capture does not guarantee improving the true multivariate-readout forecast task. We report this as a genuine, unresolved limitation rather than omit or reframe it.

Config	skill@1h	skill@3h	skill@6h	skill@12h
Matched ($\gamma_1=0.3, a=0.1$)	−1.72%	−3.82%	−4.12%	+8.89%
Reference ($\gamma_1=0.03, a=0.3$)	−4.20%	−4.59%	−1.22%	+11.15%

Table 1: IPC-matched vs. reference configuration, real reduced-sample pilot forecast skill vs. persistence.

4 Results

4.1 Full-Dataset Benchmark

Full available KORD ISD-Lite record (2011–2024, confirmed acceptable data-quality yield: 0.05–1.37% missingness per column), locked split train 2011–2020 / val 2021–2022 / test 2023–2024 (test touched exactly once). Table 2 reports skill vs. persistence at $h=1, 6$ (full results at all 48 horizons, with Diebold–Mariano tests and moving-block bootstrap CIs at every horizon, in `results/full_benchmark_test.json`).

Honest headline finding: a plain linear Ridge regression on the flattened raw window – no reservoir at all, classical or quantum – matches or beats every reservoir-computing model tested, at every horizon

Model	skill@1h (test)	skill@6h (test)
ARIMA(2,1,2)	−4890.1%	−373.0%
ARIMA (auto-order)	−626.4%	−49.9%
ESN, dimension-matched (234 units, 3 seeds)	−21.5%	+25.2%
ESN-500 (3 seeds)	+2.2%	+25.0%
Residual-ESN (3 seeds)	+13.4%	+28.6%
Null-control Ridge (no reservoir)	+14.7%	+27.0%
Null-control KRR	−280.0%	−77.7%
Residual-Ridge (no reservoir)	+14.8%	+27.0%
v4 QRC, residual (20 qubits, pilot-scale [†])	+0.09%	+0.25%
v5 QRC, residual, dissipation on (12 qubits, pilot-scale [†])	−3.51%	+5.79%

Table 2: Locked test-split headline numbers. [†]QRC rows are pilot-scale (train 2019–2021/eval 2022), reused from an earlier sprint rather than re-run on the full record, for real GPU-infrastructure time reasons (Sec. 6).

shown, confirmed on the locked test split. This is not an artifact of the pilot-scale dataset or of val-only tuning; the same conclusion holds at full 14-year scale.

4.2 Ablation: Dissipation Is the Mechanism

The central architectural claim – that engineered dissipation, not mere unitary evolution, provides the reservoir’s task-relevant memory – is tested directly by comparing the tuned configuration ($\gamma_1=0.03$) against an otherwise-identical dissipation-off diagnostic ($\gamma_1=0$), same seed, same real pilot data, 12-qubit reference config (Table 3).

γ_1	skill@1h	skill@3h	skill@6h	skill@12h (DM p)
0.03 (tuned, residual)	−3.51%	−1.36%	+5.79%	+14.39% ($p\approx 0$)
0.0 (off, residual)	−0.01%	−0.01%	−0.03%	−0.06% ($p=0.16$)

Table 3: Dissipation-on vs. off ablation, same seed and data. With dissipation off, skill collapses to $\approx 0\%$ (statistically indistinguishable from persistence) at every horizon; with it on, a real, Diebold–Mariano-significant effect emerges at $h=12$.

V1 (paradigm validation): pass. V2 (competitive verdict): fail – the same +14.4% result loses to dimension-matched ESN at 3 of 4 tested horizons (Sec. 4.1).

5 Characterization

Qubit scaling ($N \in \{4, 6, 8, 10, 12\}$, reference config, real pilot data): Memory Capacity and IPC are non-monotonic in N (dip near $N=8-10$, sharp rise at $N=12$: MC $0.85 \rightarrow 0.41 \rightarrow 0.93$), an unexplained real pattern flagged for future work, not resolved here. **Shot noise**: forecast skill is statistically indistinguishable from the exact (infinite-shot) value even at $S=1,000$ shots (bootstrap CI half-width 4.55%; Fig. 2 in `figures/sprint8_shots.png`) – a genuinely positive near-term-hardware-readiness result. **Encoding density**: sweeping injection type (RY vs. ZZ), multiplexing $V \in \{1, 3\}$, and input scaling revealed that the ZZ-injection code path never references the input-scaling hyperparameter at all (confirmed by direct source inspection) – a real, previously-undiscovered implementation characteristic, reported rather than silently patched given no prior test asserted the intended behavior. **Noise reconciliation**: depolarizing

noise on the windowed (v4) architecture’s fresh-encode step is harmful (consistent with Hou et al.’s T1-pivotal finding for encoding-stage decoherence); amplitude damping between steps in the sequential (v5) architecture is the computational resource itself (Table 3), consistent with Antoncich’s characterization of hardware-induced dissipation as regularization when the architecture is explicitly built around it – these are not contradictory findings once the mechanism is understood as architecture-dependent. The Dirac-3 hybrid stage (Sprint 7) remains pending an organizer clarification on the intended problem formulation and was not blocking this submission.

6 Limitations and Stakeholder Impact

This model does not yet beat a linear baseline. The single most important honest limitation: on this real KORD forecasting task, at both pilot and full-record scale, on the locked test split, a plain linear Ridge regression on the raw flattened window matches or exceeds every quantum and classical reservoir variant tested. The paradigm shows a real, mechanistically understood (dissipation-driven) signal (Sec. 4.2), but that signal has not yet translated into practical forecasting advantage over the simplest possible baseline.

Scope reductions, logged not hidden: the full ablation originally planned ($2 \gamma_1$ values $\times$ 2 multiplexing values $\times$ 3 seeds, 12 drives) was reduced to 2 drives (1 seed, $V=1$ only) for real time-budget reasons after two real throughput bugs were found and fixed in the GPU implementation (a $\sim 75\times$ combined speedup); the full 12-drive ablation is estimated at ~ 21 CPU/GPU-hours post-fix and was not run. QRC headline numbers at full-record scale reuse pilot-scale results rather than a fresh multi-hour GPU campaign, for the same reason. **Real bugs found and fixed, not hidden:** a PennyLane auto-queuing double-application bug in the windowed encoder (every Hadamard silently cancelled, every rotation angle doubled); two independent throughput bugs in the sequential reservoir’s GPU implementation (per-step CPU $\leftrightarrow$ GPU round-trips, then ~ 700 small GPU calls per step where a handful of vectorized gathers suffice); a missing-data-handling bug (real KORD sensor gaps left unfilled in the continuous drive sequence, corrupting the recurrent state). Each is documented with before/after numbers in the corresponding sprint report. **Stakeholder impact:** an operational forecaster choosing between this reservoir and a linear regression should, honestly, choose the linear regression today; the value delivered here is a rigorously validated negative result and a mechanistic understanding (dissipation matters) that motivates further architectural work, not a deployment-ready model.

LLM-use disclosure. Substantial portions of this project’s code, experiment scripts, and this write-up were produced with the assistance of a large language model (Claude, Anthropic) operating as a coding agent under continuous human direction and review. Every numerical claim in this paper is traceable to a `results/*.json` file generated by code in this repository; the human author(s) directed scope, reviewed real bugs and results as reported, and made the final calls on scope reductions documented in Sec. 6.

References

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